

SOUTH JERSEY, NJ, USA

WMO: 724074

Lat: 39.941N

Lon: 74.841W

Elev: 49

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93780

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	15.9	-5.6	4.1	14.1	-0.4	5.4	19.2	21.9	36.3	19.9	32.3	6.8	290	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	92.5	74.5	90.2	73.6	87.8	72.6	77.7	87.1	76.4	84.9	75.2	82.9	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	132.1	81.0	73.7	125.8	79.7	72.8	121.9	78.9	41.0	86.8	39.8	84.7	38.6	83.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.4	14.0		6.0	96.3	4.7	3.1	2.7	98.5	-0.1	100.3	-2.7	102.1	-6.1	104.3
				5.0	80.4	4.7	1.2	1.7	81.3	-1.0	82.0	-3.7	82.6	-7.0	83.5
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	32.4	34.8	42.6	53.3	62.9	71.5	76.2	74.5	67.9	56.5	46.3	38.2	(6)
(7)		DBStd	17.12	10.31	9.18	9.01	8.50	7.70	6.41	5.07	5.14	6.47	8.08	8.21	9.12	(7)
(8)		HDD50	1887	552	435	263	56	2	0	0	0	28	168	383	(8)	
(9)		HDD65	4791	1011	847	693	367	135	17	0	1	43	285	562	830	(9)
(10)		CDD50	3664	7	8	35	154	402	645	813	760	536	229	57	18	(10)
(11)		CDD65	1093	0	0	0	15	70	212	348	297	130	21	0	0	(11)
(12)		CDH74	9902	0	0	9	189	707	1961	3335	2560	1011	128	2	0	(12)
(13)		CDH80	3584	0	0	0	51	218	707	1377	927	285	19	0	0	(13)
(14)	Wind	WSAvg	5.4	6.6	6.7	7.3	6.6	5.0	4.6	3.9	3.4	3.9	4.8	5.6	6.0	(14)
(15)	Precipitation	PrecAvg	45.4	3.7	2.5	4.2	3.7	3.6	3.8	4.5	4.8	3.7	3.7	3.2	4.1	(15)
(16)		PrecMax	57.2	6.5	5.0	7.7	8.9	6.8	8.8	10.5	13.0	7.5	10.4	6.5	8.6	(16)
(17)		PrecMin	34.2	1.5	0.5	1.0	1.0	1.2	1.4	1.7	1.2	0.4	0.6	0.9	1.1	(17)
(18)		PrecStd	5.8	1.3	1.1	1.7	1.7	1.3	2.0	1.9	2.5	2.0	2.1	1.4	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.0	67.5	74.6	86.3	90.1	94.3	97.2	95.1	91.2	82.4	72.3	68.2	(19)
(20)		0.4%	MCWB	59.3	55.7	60.0	65.6	71.2	74.7	75.5	75.9	73.9	69.3	63.1	62.5	(20)
(21)		2%	DB	59.5	60.8	68.1	79.7	86.2	90.4	93.1	91.0	86.9	78.0	67.8	62.5	(21)
(22)		2%	MCWB	55.1	52.7	56.3	62.2	69.9	73.5	74.8	74.8	72.0	67.6	59.6	58.2	(22)
(23)		5%	DB	53.6	54.8	63.0	73.9	81.7	87.6	90.4	88.1	82.9	73.7	63.9	57.9	(23)
(24)		5%	MCWB	48.1	47.8	52.9	59.9	67.6	72.5	73.9	73.1	70.3	65.2	57.5	53.7	(24)
(25)		10%	DB	47.8	48.8	57.5	69.2	77.6	84.0	87.9	85.2	80.1	70.3	60.6	52.8	(25)
(26)		10%	MCWB	42.7	42.4	49.6	57.7	65.5	70.2	73.2	71.8	69.1	63.1	54.8	48.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.7	60.8	62.4	68.8	74.0	77.4	79.7	79.5	77.2	73.4	65.4	63.5	(27)
(28)		0.4%	MCDWB	63.5	65.5	71.0	79.6	85.0	88.4	91.4	90.1	85.1	77.5	69.5	65.7	(28)
(29)		2%	WB	55.5	54.9	58.4	65.3	71.9	75.8	77.7	77.8	75.4	70.3	62.5	59.1	(29)
(30)		2%	MCDWB	59.0	58.8	65.5	75.3	82.5	85.7	87.7	86.6	81.5	75.3	66.0	61.8	(30)
(31)		5%	WB	48.9	47.9	54.6	62.5	69.9	74.5	76.4	76.3	73.6	67.6	59.1	54.7	(31)
(32)		5%	MCDWB	52.4	53.9	60.6	71.3	79.1	83.5	85.5	83.7	78.7	72.0	62.9	57.6	(32)
(33)		10%	WB	43.7	43.1	50.2	59.3	67.7	72.9	75.2	74.9	71.9	64.3	55.7	49.1	(33)
(34)		10%	MCDWB	47.1	48.4	57.4	67.9	75.4	81.0	83.5	81.2	77.4	68.9	59.4	52.0	(34)
(35)	Mean Daily Temperature Range		MDBR	15.4	16.7	18.3	20.8	20.4	19.5	19.2	18.9	19.9	18.8	18.5	15.5	(35)
(36)		5% DB	MCDBR	23.5	24.9	26.7	28.4	25.5	22.9	22.1	21.4	23.2	22.7	23.0	19.9	(36)
(37)		5% DB	MCWBR	19.4	19.0	17.2	14.5	12.1	9.4	7.8	8.1	10.5	13.0	16.1	18.0	(37)
(38)		5% WB	MCDWR	21.9	23.5	24.2	24.2	22.4	20.0	18.7	17.5	17.9	17.9	18.8	19.3	(38)
(39)		5% WB	MCWBR	19.8	19.7	18.0	14.8	11.7	9.5	7.8	7.8	9.6	12.4	15.7	18.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.321	0.360	0.391	0.438	0.502	0.503	0.495	0.409	0.361	0.329	0.310	(40)	
(41)		taud	2.465	2.437	2.385	2.322	2.222	2.085	2.097	2.105	2.318	2.461	2.509	2.522	(41)	
(42)		Ebn at Noon	271	285	284	280	268	250	249	247	263	266	263	264	(42)	
(43)		Edh at Noon	25	30	35	39	44	51	50	49	37	29	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	615	903	1197	1546	1716	1882	1918	1666	1381	984	700	536	(44)	
(45)		RadStd	54	98	97	133	201	156	91	108	143	100	62	53	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (71 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+126

Nomenclature: See separate page